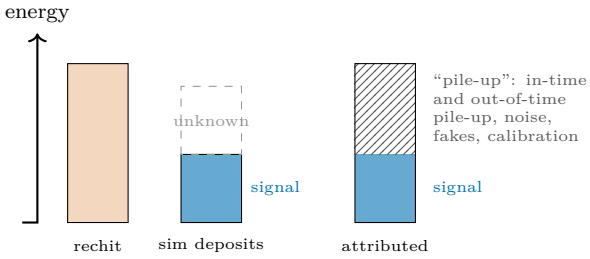
## premixed sample (`run3_rawsim`)

the simulation knows the hard-scatter particles only




**cell**: cap = $\min(E_{\text{rec}},\, s\,E_{\text{sim}}^{\text{signal}})$, shared among the signal particles $\propto$ deposits; the rest is unattributed

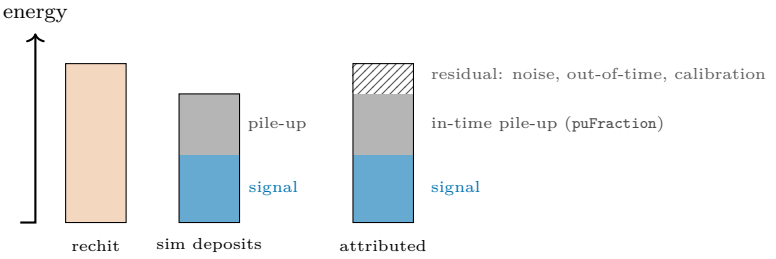
**track**: TrackingParticle of the hard-scatter event $\Rightarrow$ signal; none $\Rightarrow$ nothing (every pile-up track lands here)

**candidate**: `signalFraction`; $1-$ `signalFraction` = pile-up + noise + fakes + calibration, in one number

**jet**: `signalPtRatio`; `ptRatio` $-$ `signalPtRatio` = the unattributed part, split by candidate type only (section 4.6)

## classically mixed sample (`run3_mix`)

the pile-up particles and their deposits are known too




**cell**: cap = $\min(E_{\text{rec}},\, s\,(E_{\text{sim}}^{\text{signal}} + E_{\text{sim}}^{\text{pile-up}}))$, shared among signal and pile-up particles $\propto$ deposits; the rest is the residual

**track**: TrackingParticle of the hard-scatter event $\Rightarrow$ signal; of a pile-up interaction $\Rightarrow$ pile-up; none $\Rightarrow$ residual

**candidate**: `signalFraction`, `puFraction`, residual = $1-$ both

**jet**: the same `signalPtRatio`; the unattributed part split into true in-time pile-up by origin and residual; the pile-up rungs `SimPFPUJet`, `SimPFPUCHSJet`, `SimCaloCellPUJet`

**the signal part is defined identically on both samples**: same cap for the signal, same shares, same `signalPtRatio` and shared-energy response.
The ladders agree within a few percent; the mixed sample adds what the unattributed part is made of.